

AP Precalculus: 2026–2027

Unit 4: Functions Involving Parameters, Vectors, and Matrices Lesson 4.5: Implicitly Defined Functions

Primary Objective: Students will graph equations in two variables by finding solutions and identifying one or more implicit functions, determine how the variables co-vary by examining the sign of their ratio of change, and locate where the rate of change is zero or undefined and interpret this geometrically. (Big Idea: Functions)

Priority Ideas & Skills

AP Skill 2.B — Construct equivalent representations

AP Skill 3.A — Describe characteristics of a function

Learning Objectives:

- **LO 4.5.A:** Construct a graph of an equation involving two variables.
- **LO 4.5.B:** Determine how the two quantities related in an implicitly defined function vary together.

Key Understandings (EKs):

- An equation in two variables can implicitly describe one or more functions. (EK 4.5.A.1)
- A graph can be built by substituting values and solving for the other variable. (EK 4.5.A.2)
- Solving for one variable yields an explicit function whose graph is part or all of the equation's graph. (EK 4.5.A.3)
- If $\Delta y/\Delta x$ is positive, the variables change in the same direction; if negative, they change in opposite directions. (EK 4.5.B.1)
- $\Delta y/\Delta x = 0$ signals a horizontal interval; undefined signals a vertical interval. (EK 4.5.B.2)

Vocabulary, Concepts, & Theorems

Implicit equation

An equation in two variables where neither variable is isolated; it may describe one or more functions.

Explicitly defined function

A function of the form $y = f(x)$, where y is solved for in terms of x .

Co-variation

The way two quantities change together; characterized by the sign and magnitude of $\Delta y/\Delta x$.

Rate of change

$\Delta y/\Delta x$: the ratio of the change in y to the change in x between two nearby points. Positive $\Rightarrow$ same direction; negative $\Rightarrow$ opposite directions.

Activate Prior Knowledge & Spiral Review

Students should already be able to:

- Solve a two-variable equation for one variable algebraically.
- Evaluate $y = f(x)$ for specific x values and plot (x, y) pairs.
- Compute $\Delta y / \Delta x$ from a table of values.
- Use the square-root property ($y^2 = k \Rightarrow y = \pm\sqrt{k}$).

Warm-Up covers: Solving $x^2 + y^2 = 100$ for y ; reading a ratio from a table; finding y -values for $y^2 = x + 4$.

Hook

“The equation $x^2 + y^2 = 25$ is not a function — but it describes a circle. Can we still talk about how x and y change together? Can we split it into functions?”

Ask students: Is $x^2 + y^2 = 25$ a function? Why or why not? (No — fails the vertical-line test.) What if we only look at the top half? Guide them to see that each semicircle *is* a function.

Lesson — Day 1: Graphing Implicit Equations

Step 1 — What is an implicit equation?

An equation like $x^2 + y^2 = 25$ involves two variables but neither is isolated. We call it **implicit** because functions are hidden inside it.

Pose: “How could we graph this?”

Guide: Substitute specific x values and solve for y .

Example: $x = 3$ in $x^2 + y^2 = 25$:

$$9 + y^2 = 25 \implies y^2 = 16 \implies y = \pm 4$$

Two points, $(3, 4)$ and $(3, -4)$, for one x -value.

Step 2 — Identifying the explicit pieces.

Solve $x^2 + y^2 = 25$ for y :

$$y = \pm\sqrt{25 - x^2}$$

Two explicit functions on $[-5, 5]$:

- $f_1(x) = \sqrt{25 - x^2}$ (upper semicircle)
- $f_2(x) = -\sqrt{25 - x^2}$ (lower semicircle)

Step 3 — Table for $x^2 + y^2 = 25$.

x	y (upper)	y (lower)
−5	0	0
−3	4	−4
0	5	−5
3	4	−4
5	0	0

Have students verify each row by substitution.

Step 4 — Second example: $xy = 4$.

Solve: $y = 4/x$. One explicit function (two branches, no repeated y).

x	y
1	4
2	2
4	1
−1	−4
−2	−2

Pose: “As x increases from 1 to 4, what happens to y ?” (Decreases from 4 to 1.)

Lesson (cont.) — Day 2: Co-variation

Step 5 — Co-variation sign.

Pick two nearby points on the upper semicircle of $x^2 + y^2 = 25$: (3, 4) and (4, 3).

$$\frac{\Delta y}{\Delta x} = \frac{3 - 4}{4 - 3} = -1$$

Negative $\Rightarrow$ as x increases, y decreases. **Opposite directions.**

Pose: “What would a *positive* ratio mean?”

(Both variables increasing, or both decreasing together.)

Step 6 — Where is the ratio zero or undefined?

- Near (0, 5): $\Delta y/\Delta x \rightarrow 0$. y barely changes — **horizontal** behavior at the top.
- Near (5, 0): $\Delta y/\Delta x$ becomes very large (undefined limit) — **vertical** behavior at the right end.

Summary:

Ratio	Interpretation
Positive	Same direction (both $\uparrow$ or both $\downarrow$)
Negative	Opposite directions
Zero	y momentarily constant (horizontal)
Undefined	x momentarily constant (vertical)

$xy = 4$ **example:**

Points (2, 2) and (3, 4/3):

$$\frac{\Delta y}{\Delta x} = \frac{4/3 - 2}{3 - 2} = -\frac{2}{3}$$

Negative: as x increases, y decreases. Always true when $xy = k > 0$.

Active Monitoring

Circulate and check for:

- Students who find only one y -value when two exist (e.g., forgetting $y = -4$ when $y^2 = 16$). Prompt: “ $y^2 = 16$ — what are ALL values of y ?”
- Students who compute $\Delta x/\Delta y$ instead of $\Delta y/\Delta x$.
- Students who confuse the sign of the ratio with the sign of the y -values themselves.
- Students who cannot identify which piece of the circle is a function; prompt with the vertical-line test.

Cold-call prompts:

- “If $\Delta y/\Delta x$ is negative, what does that tell us about co-variation?”
- “How many explicit functions does $x^2 + y^2 = 25$ define? Name them.”
- “At which points on the circle is the rate of change equal to zero?”

Group Work & Differentiation

Tier R — Remediate

For $x^2 + y^2 = 16$, find four solutions by substituting $x = -4, 0, 4$ and one other value. Determine whether y is increasing or decreasing as x increases from -4 to 0 on the upper half.

Tier A — Approaching Proficiency

For $x^2 + 4y^2 = 16$, find solutions for $x = -4, -2, 0, 2, 4$, identify the two explicit functions, and determine where $\Delta y/\Delta x$ changes sign.

Tier E — Extension

For $x^2 - y^2 = 9$, identify two explicit functions and state their domains. Describe co-variation in each quadrant. What happens at $x = 3$ and $x = -3$?

Individual Work & Assessment

Exit Ticket (2 items):

- The equation $4x^2 + 9y^2 = 36$ (an ellipse). For $x = 3$, find the y -values. For $x = 0$, find the y -values.
- Using (3, 0) and (0, 2) from $4x^2 + 9y^2 = 36$, determine whether co-variation is positive or negative as we move from (3, 0) toward (0, 2) along the upper half of the ellipse.

Assessment note: Sort exit tickets by whether students (a) found both y -values and (b) correctly identified the sign of co-variation. Use results to form Tier R/A/E groups for Day 2 re-teach if needed.

Reinforcement & Extension

Homework: 5–6 problems including solving for y from an implicit equation, finding ordered-pair solutions, identifying explicit function pieces, determining co-variation sign, a contextual budget-constraint problem, and one AP-style MC item.

Extension: Explore $x^2 - y^2 = 9$ (a hyperbola). Identify its two branches, describe co-variation in each branch, and find where the rate of change is zero or undefined.

Preview of Lesson 4.6: We will extend implicit equations to parametric representations, where both x and y are expressed as functions of a third variable t .